

JAVE N. ABEJUELA

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EXPERIENCE/PROJECTS

EMVIEM TELECOMMUNICATION SERVICES (Student Intern)

- Drafted detailed "As-Built" and "For Implementation" technical plans using AutoCAD for the relocation and installation of Network Access Points (NAP) within an Optical Distribution Network (ODN).
- Assisted field technicians in fiber optic splicing and performed signal testing using an Optical Power Meter (OPM) to ensure network integrity and minimize decibel (dB) loss during maintenance.
- Collaborated on OSP (Outside Plant) design by mapping infrastructure routes and updating distribution terminal records to reflect real-world network modifications.

TOMATOOL: AN INTERNET-OF-THINGS SENSING-BASED DEVICE WITH MACHINE LEARNING ALGORITHMS AND SMART CROP CALENDAR FOR TOMATOES

- Deployed and integrated a trained predictive model into a live system, ensuring accurate data flow from hardware sensors to the AI inference engine for yield or price forecasting.
- Developed a comprehensive IoT data pipeline using sensor nodes to collect real-time environmental data, implementing a server-side storage system for machine learning processing and live visual monitoring via a web dashboard.
- Conducted rigorous testing and calibration of sensors to ensure data precision, and utilized 3D modeling software to design custom enclosures for the physical sensor nodes, optimizing them for field deployment.

IMAGE DETECTION OF CAPACITIVE, INDUCTIVE, AND RESISTIVE ELECTRONIC DEVICES

- Curated a specialized dataset of 1,000 high-resolution images of capacitive, inductive, and resistive components, performing manual annotation and labeling to ensure high ground-truth accuracy for object detection training.
- Optimized model generalization by implementing image preprocessing procedures and augmentation techniques.

EDUCATION

Polytechnic University of the Philippines (2022-Present)
(Bachelor of Science in Electronics Engineering)

Malvar Senior High School (2020-2022)
(STEM)

Malvar School of Arts and Trade (2018-2020)
(Technical Vocational School Major in Electronics Technology)

Certifications:

- Python 101 for Data Science, IBM Cognitive Class (March 2025)
- Understanding LLMs and Basic Prompting Techniques, CodeSignal (April 2025)
- Data Science and Analytics, HP life (February 2026)
- Ai for Business Professionals, HP life (February 2026)

Organization:

- Organization of Electronics Engineering Students (2022 – Present)
- Institute of Electronics Engineers of the Philippines Batangas Student Chapter (2022 – Present)

SKILLS

TECHNICAL SKILLS

- **Engineering & Design:** AutoCAD (ODN/OSP Design), MATLAB, 3D Modeling, Circuit Troubleshooting.
- **AI & Data Science:** Machine Learning Deployment, Data Analysis, Dataset Annotation & Augmentation, Jupyter Notebook.
- **Software & Tools:** VS Code, Git/GitHub, Arduino IDE, AI-Assisted Development (Prompt Engineering).
- **Hardware & Testing:** ESP32/Raspberry Pi Integration, Sensor Calibration, Fiber Optic Splicing, Optical Power Meter (OPM).

SOFT SKILLS

- Collaborative Teamwork
- Effective Communication & Active Listening
- Adaptive Problem Solving
- Emotional Intelligence